

Topic 2:

Components of a RCT

PUBH 526

Design and Analysis of Randomized Trials in Public Health

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Reading

- Friedman, Furberg, DeMets, Reboussin, and Granger. (2015) Fundamentals of Clinical Trials, Fifth Edition.

Iron During Pregnancy

- Anemia is common in pregnancy and increases the risk of adverse outcomes
- Iron deficiency is a leading cause of anemia in sub-Saharan Africa
- Iron supplementation is the universal standard of care during pregnancy
 - Regardless of iron/anemia status
- However, a large trial among children in Tanzania raised concerns about the safety of iron supplementation in malaria-endemic regions (Sazawal et al. 2006)
 - Study was stopped early because children receiving iron were more likely to die or be hospitalized



Iron Supplementation in Iron-^{Sufficient}Replete and Nonanemic Pregnant Women in Tanzania

A Randomized Clinical Trial

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- In malaria-endemic regions:
 - Iron supplementation can help in cases of iron deficiency or anemia
 - In other cases, could iron do more harm than good?
- Objective: To evaluate the safety and efficacy of iron supplementation during pregnancy in a malaria-endemic region
- Randomized, double-blind, placebo-controlled clinical trial among pregnant women in Dar es Salaam, Tanzania

TRIAL REGISTRATION [clinicaltrials.gov](https://clinicaltrials.gov/ct2/show/study/NCT01119612) Identifier: NCT01119612

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Specific Aims

- Based on existing knowledge or theory
- Stated in advance
 - Prevents:
 - Avoidance of non-significant or unexpected results
 - Fishing for statistical significance
- Statistical **power** calculations are done on primary aims
 $\phi(\text{Reject } H_0 \mid H_1 \text{ true})$

Iron Supplement = IS

Primary Aims

(A)
Safety
of IS

H_0 : 'IS' does not change incidence or density of placental malaria compared to placebo. ^{placebo} ^{μ iron}
 H_A : 'IS' does increase ↑ incidence or density of placental malaria compared to placebo.

(B)
Efficacy
(Hb)

H_0 : 'IS' does not change maternal Hb level compared to placebo. ^{μ iron}
 H_A : 'IS' does increase ↑ maternal Hb level compared to placebo.

- Null hypotheses?

- Directionality of “safety” and “efficacy”? ^{one-sided[†]}
 ↪ might ↑ malaria risk ↪ Improvement (beneficial)

- (A) To determine the **safety** of iron supplementation among pregnant Tanzanian women, as measured by (a) the incidence of placental malaria, and (b) placental malaria parasite density, compared to placebo.
- (B) To determine the **efficacy** of iron supplementation among pregnant Tanzanian women, as measured by maternal hemoglobin concentrations, compared to placebo.
- (C) To determine the **efficacy** of iron supplementation among pregnant Tanzanian women, as measured by infant birth weight, compared to placebo.

(C)
Efficacy
(weight)

H_0 : 'IS' does not change infant birth weight compared to placebo. ^{μ iron}
 H_A : 'IS' does increase ↑ infant birth weight compared to placebo.

Secondary Aims

- Compare Primary Aims 2&3 to Secondary Aims 4&5

4. To determine the **efficacy** of iron supplementation among pregnant Tanzanian women, on the **incidence of maternal anemia**, compared to placebo.
5. To determine the **efficacy** of iron supplementation among pregnant Tanzanian women, on the **incidence of low birth weight**, compared to placebo.
6. To determine the **efficacy** of iron supplementation among pregnant Tanzanian women, on the **incidence of maternal malaria infection**, compared to placebo.

- Type I error: Reject H_0 | H_0 true
- Type II error: Do not reject H_0 | H_A true
- Power: probability of finding something when it's actually there, i.e., $P(\text{reject } H_0 | H_A)$

		REALITY	
		H_0 Treatments are NOT different	H_A Treatments ARE different
POSSIBLE CONCLUSIONS	Do not reject H_0 We conclude treatments are NOT different from each other	Correct decision (cell a) <i>match</i>	Type II error (probability = β) (cell b) <i>✗ Fake Neg.</i>
	Reject H_0 We conclude treatments ARE different from each other	Type I error (probability = α) (cell c) <i>✗ False Pos.</i>	Correct decision (cell d)

Methods

- Placebo-controlled clinical trial
- Randomized
- Double-blind *Ex): random codes for Trt v.s. Placebo*
- Among pregnant women presenting for antenatal care in Dar
- Inclusion and exclusion criteria
 - Iron-replete, nonanemic women were eligible if they were uninfected with human immunodeficiency virus, primigravidae or secundigravidae, and at or before 27 weeks of gestation *Ex): Exclude women who are giving birth within 3 weeks → not long enough to show true effect*
 - All terms must be well defined in advance
 - Defines the study population
 - Changes to these criteria change the population to which you can generalize

Methods

- Screening of 21,316 women continued until the **target enrollment of 1500** was reached
- Analyses followed the **intent-to-treat principle** and included all randomized participants
 - Analyze study participants according to their treatment assignment
 - Assume women took 100% of their iron pills OR 100% of their placebo pills, and analyze effect of iron during pregnancy by comparing the two groups of women
 - It's very likely that some participants don't take 100% of their pills. What kind of **bias** does this create?
 - **Bias:** ^{bias toward the H₀, underestimation} an estimate systematically **deviates from the true value** (e.g., the effect of iron on an outcome that you calculate from your data systematically **overestimates or underestimates** the true effect of iron compared with the placebo)
 - In this particular example, this is **"bias towards the null"**. ✓
Underestimating the effect of a treatment is **conservative** and usually preferred to overestimating. ^{◦ Avoid unnecessary invest in money/resources} ^{◦ Prefer improvement over "scaling up".}

Methods

- Alternative analyses are possible:
 - Analyze only women who took all or most of their pills
 - Analyze women who took all/most of the iron pills vs. all women in the control group
 - Include % of pills taken in the analysis (look for a dose-response relationship)

- So, why do **intention-to-treat?**

include all patients
regardless of Trt.
adherence →

ITT analysis is commonly used in:

- Randomized controlled trials (RCTs)
- Observational studies where treatment assignment is not randomized, but there is an intention to treat all participants

Example:

Imagine a study where patients are randomized to receive either a new medication or a placebo. If some patients in the medication group do not take the medication as prescribed, ITT analysis would still include these patients in the medication group when evaluating the treatment effect.

Advantages:

- Provides an unbiased estimate of treatment effect
- Maintains the integrity of the randomization process
- Generalizable to real-world practice, where patients may not always adhere to treatment

Disadvantages:

- May underestimate the true treatment effect if non-adherence is high
- Does not provide information about the effectiveness of the treatment in patients who adhere to the protocol

Data Collection

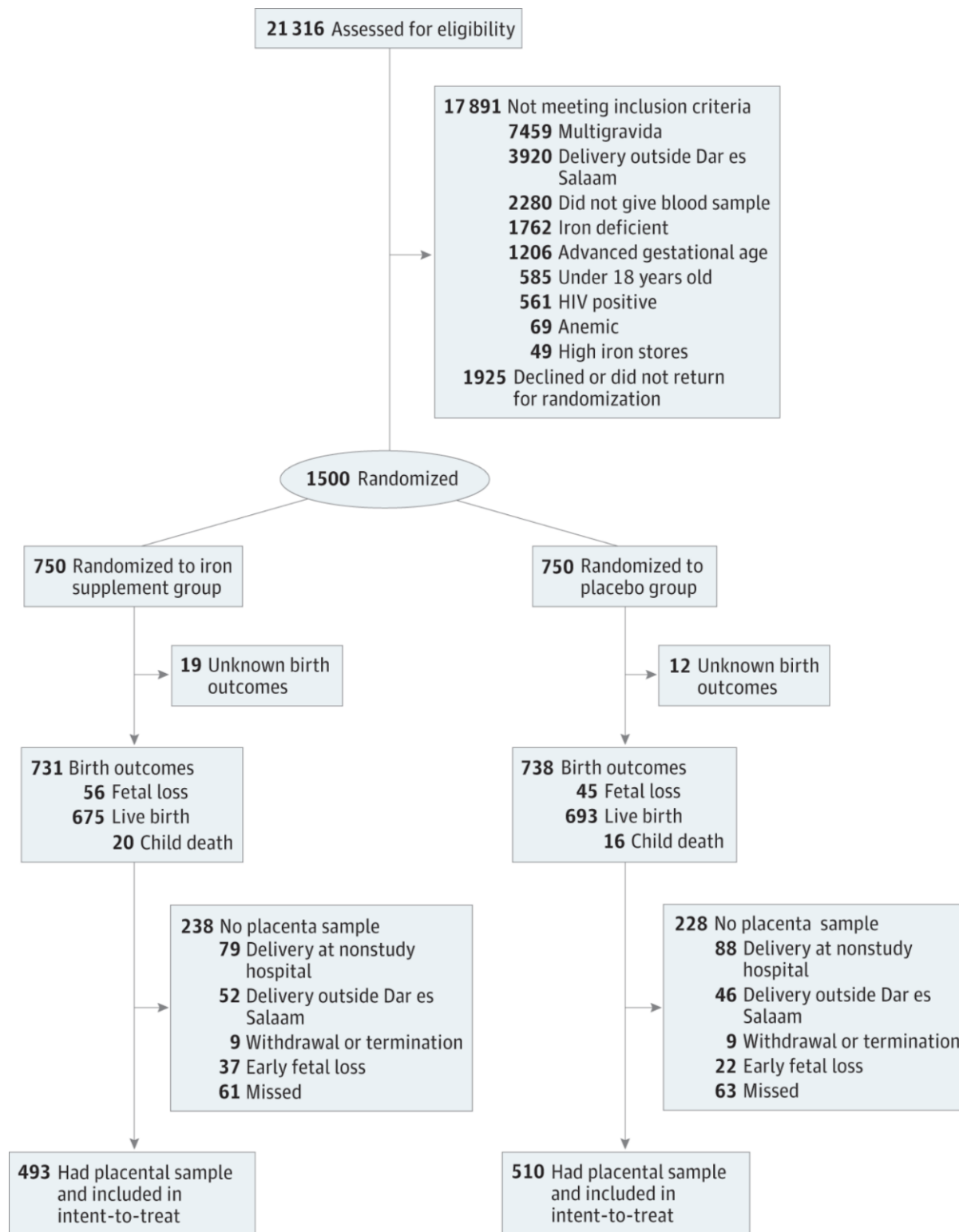
- Every participant is on her own timeline
- One participant becomes two at delivery!

	Screening	Randomiza tion	Monthly	30 Weeks	Delivery	6-wk Postpartum
Forms						
Screening Form	X					
Randomization		X				
Background Form		X				
Clinical Forms		X	X	X	X	X
Food Frequency Form		X			X-36 wk	
Delivery Form					X	
Newborn Form					X	
Laboratory Tests						
Serum ferritin		X			X*	
C-Reactive Protein		X			X*	
Hepcidin		X*			X*	
Transferrin Receptor		X*			X*	
Hemogram		X		X	X	
Blood Smear for parasites		X		X	X	X
Placental and Cord					X	
Blood Smears					X	
Stool General		X			X	

* These tests will be done on a sample of 300 individuals.

Reporting

- CONSORT Statement
- <https://www.consort-spirit.org/>
- Review checklist
- Review flowchart



- Are these a concern?
- Are they only a concern if there are differences between arms?

Table 1. Participant Characteristics at Baseline^a

Characteristic	Iron Group (n = 750)	Placebo Group (n = 750)
Age, mean (SD), y	23.7 (4.1)	24.1 (4.2)
Primigravida	451 (60.1)	415 (55.3)
Gestational age at randomization, mean (SD), wk	18.2 (4.3)	18.2 (4.4)
Educational level, y		
0-4	18 (2.5)	14 (1.9)
5-7	404 (55.0)	389 (53.6)
8-11	211 (28.8)	202 (27.8)
≥12	101 (13.8)	121 (16.7)
Marital status		
Married or cohabitating	595 (79.7)	600 (80.5)
Other	152 (20.4)	145 (19.5)
Employment status		
Skilled	140 (18.7)	149 (19.9)
Unskilled or informal	226 (30.1)	226 (30.1)
Housewife or unemployed	362 (48.3)	348 (46.4)
Other	22 (2.9)	27 (3.6)
Housing type		
Own	139 (18.6)	143 (19.1)
Rent	534 (71.5)	534 (71.3)
Other	74 (9.9)	72 (9.6)
Roof		
Metal	730 (97.7)	734 (98.0)
Other	17 (2.3)	15 (2.0)
Floor		
Concrete	701 (93.5)	704 (93.9)
Tile	43 (5.7)	41 (5.5)
Dirt or wood	6 (0.8)	5 (0.7)
Malaria interventions		
Uses spray, coils, or ITN nightly	668 (90.6)	686 (92.7)
Has ITN	663 (88.5)	666 (88.9)
Uses ITN	628 (86.2)	638 (87.2)
BMI at enrollment, mean (SD)	24.5 (4.4)	24.6 (4.7)
BMI category at enrollment		
<18.5	32 (4.3)	36 (4.8)
18.5-24.9	434 (58.3)	415 (55.6)
≥25	278 (37.4)	295 (39.5)
Baseline MUAC, mean (SD), mm	26.5 (3.6)	26.5 (3.6)

Table 2. Effect of Iron Supplementation vs Placebo on Maternal Outcomes^a

Outcome	Iron Group		Placebo Group		Relative Risk (95% CI)	P Value
	Finding	No. of Participants or Events	Finding	No. of Participants or Events		
Placental malaria ^b						
Microscopic	11 (2.3)	471	10 (2.1)	487	1.14 (0.49-2.65)	.77
Submicroscopic	26 (5.3)	486	24 (4.8)	501	1.12 (0.65-1.92)	.69
Any placental malaria	33 (6.7)	493	33 (6.5)	510	1.03 (0.65-1.65)	.89
Placental weight, mean (SD), g	458 (140)	443	456 (144)	467	NA	.97
Maternal death	2 (0.3)	750	3 (0.4)	750	0.67 (0.11-3.98)	>.99
Maternal hospitalizations	20 (0.4)	4380	36 (0.8)	4474	0.57 (0.32-1.02)	.06
Malaria	2 (0)	4380	7 (0.2)	4474	0.29 (0.06-1.40)	.12
Abdominal pain	3 (0.1)	4380	4 (0.1)	4474	0.80 (0.17-3.40)	.73
Vomiting	1 (0)	4380	1 (0)	4474	1.02 (0.06-16.30)	.99
Other	14 (0.3)	4380	24 (0.5)	4474	0.60 (0.31-1.15)	.12
Potential adverse effects ever reported ^c	227 (30.3)	750	226 (30.1)	750	1.00 (0.86-1.17)	>.99
Ever temporarily discontinued regimen because of perceived adverse effects	25 (3.3)	750	30 (4.0)	750	0.83 (0.49-1.40)	.58

Abbreviation: NA, not applicable.

^a Data are presented as number (percentage) of participants unless otherwise indicated.

^b Microscopic infections were detected by placental histopathologic analysis.

Submicroscopic infections were detected by quantitative polymerase chain reaction on maternal placental tissue. Five participants (4 in the iron group, 1 in the placebo group) had both microscopic and submicroscopic infections.

^c Nausea, vomiting, or diarrhea reported at monthly visit.

Table 3. Effect of Iron Supplementation vs Placebo on Maternal Hematologic and Iron Status^a

	Iron Group		Placebo Group		Relative Risk (95% CI)	P Value
Outcome	Finding	No. of Participants or Events	Finding	No. of Participants or Events		
Hemoglobin, mean (SD), g/dL						
Randomization	11.7 (1.4)	738	11.7 (1.3)	741	NA	.98
Delivery	11.8 (2.0)	481	10.9 (1.9)	501	NA	<.001
Mean change in hemoglobin	0.1 (2.2)	473	−0.7 (2.0)	495	NA	<.001
Anemia at delivery ^b	144 (29.9)	481	248 (49.5)	501	0.60 (0.51-0.71)	<.001
Severe maternal anemia at delivery ^c	23 (4.8)	481	35 (7.0)	501	0.68 (0.41-1.14)	.14
Ferritin, µg/L						
Randomization	40.9 (33.3)	741	40.7 (31.4)	742	NA	>.99
Delivery	92.5 (171.1)	483	54.1 (90.3)	509	NA	<.001
Mean change in ferritin	41.3 (295.6)	476	11.3 (92.1)	503	NA	<.001
Iron deficiency						
At delivery ^d	33 (6.8)	483	73 (14.3)	509	0.48 (0.32-0.70)	.001
At delivery (CRP adjusted) ^e	227 (47.3)	480	379 (75.3)	503	0.63 (0.56-0.70)	<.001
Anemia at delivery ^{b,d}	14 (3.0)	464	43 (8.8)	491	0.34 (0.19-0.62)	<.001
Anemia at delivery (CRP adjusted) ^{c,e}	74 (16.1)	461	191 (39.4)	485	0.41 (0.32-0.52)	<.001

Abbreviations: CRP, C-reactive protein; NA, not applicable.

SI conversion factors: To convert hemoglobin to grams per liter, multiply by 10; ferritin to picomoles per liter, multiply by 2.247.

^a Data are presented as number (percentage) of participants unless otherwise indicated.

^b Hemoglobin level less than 11 g/dL.

^c Hemoglobin level less than 8.5 g/dL.

^d Serum ferritin level less than 12 µg/L.

^e Serum ferritin level less than 70 µg/L and CRP level greater than 8.2 µg/mL or serum ferritin level less than 30 µg/L and CRP level of less than or equal to 8.2 µg/mL.

Table 4. Effect of Iron Supplementation vs Placebo on Perinatal Outcomes^a

Outcome	Iron Group		Placebo Group		Relative Risk (95% CI)	P Value
	Finding	No. of Participants or Events	Finding	No. of Participants or Events		
Fetal loss ^b	56 (7.7)	731	45 (6.1)	738	1.26 (0.86-1.92)	.24
Infant mortality at 6 wk ^c	20 (3.0)	675	16 (2.3)	692	1.28 (0.67-2.45)	.45
Birth weight, mean (SD), g	3155 (545)	672	3137 (519)	691		.89
Birth weight group, g						
<2500	45 (6.7)	672	51 (7.4)	691	0.91 (0.62-1.34)	.62
<2000	15 (2.2)	672	14 (2.0)	691	1.10 (0.54-2.26)	.79
Gestational age at delivery, mean (SD), wk	39.3 (2.5)	665	39.2 (2.7)	685		.37
Preterm birth, wk ^b						
<37	100 (15.0)	665	113 (16.5)	685	0.91 (0.71-1.17)	.46
<34	21 (3.2)	665	27 (3.9)	685	0.80 (0.46-1.40)	.44
Low birth weight and preterm birth ^{d,e}	27 (4.1)	665	21 (3.1)	685	1.32 (0.76-2.32)	.32
Low birth weight and term birth ^{d,e}	18 (2.7)	665	30 (4.4)	685	0.62 (0.35-1.10)	.10
Small for gestational age (Alexander et al standard) ^{d,f}	121 (18.0)	672	125 (18.1)	691	1.00 (0.79-1.25)	.97
Small for gestational age (INTERGROWTH-21st standard) ^g	101 (15.8)	641	105 (16.2)	648	0.97 (0.76-1.25)	.83

Abbreviation: INTERGROWTH-21st, International Fetal and Newborn Growth Consortium for the 21st Century.

^a Data are presented as number (percentage) of participants unless otherwise indicated.

^b Stillbirth and miscarriage.

^c Live birth only.

^d Restricted to births at 21 to 44 weeks of gestation.

^e Low birth weight was defined as less than 2500 g.

^f Small for gestational age was defined as birth weight below the 10th percentile for gestational age using the standard of Alexander et al.¹⁸

^g Small for gestational age was defined as birth weight below the 10th percentile for gestational age using the INTERGROWTH-21st standard,¹⁹ restricted to births at 30 to 42 weeks of gestation.

What Can We Conclude?

- Prenatal iron supplementation among iron-replete, nonanemic women was not associated with an increased risk of placental malaria or other adverse events in the context of good malaria control
 - Participants receiving supplementation had improved hematologic and iron status at delivery compared with the placebo group
- These findings provide support for continued administration of iron during pregnancy in malaria-endemic regions

Limitations?

- Lower than expected incidence of malaria reduces statistical power
 - Good preventative measures for malaria had become more common
 - Implications for external validity
- Loss to follow-up and missing data, particularly for placental malaria outcomes
 - Some concerns can be explored statistically

Power Calculations

Statistical Power for Incidence of Placental Malaria				
Incidence in Placebo Arm	Effect of Intervention on the Incidence			
	HR=0.75	HR=0.70	HR=0.65	HR=0.60
20%	48%	65%	80%	90%
25%	57%	74%	87%	95%
30%	64%	80%	92%	98%

The study will have good power to detect a 35% effect size, for placental malaria rates as low as 20% in the placebo arm. If the placental malaria rates are larger (25% or 30%), the study will have good power to detect a 30% effect size and excellent power to detect a 40% effect size.

- Hazard ratio (HR) is the ratio of the hazard rates with vs. without the intervention
- HR=0.70: placental malaria occurs at a 30% lower rate in the placebo arm than in the iron arm

Power Calculations

Statistical Power for Continuous Variables							
Variable (Units)	Mean	SD	Minimal effect size for 80% power	Power for 5% change	Power for 10% change	Power for 15% change	Power for 20% change
Placental malaria density (parasites/ μ L)	9000	5000	8.5%	38%	91%	>99%	>99%
Maternal hemoglobin concentration (g/dL)	11.4	3.8	5.0%	79%	>99%	>99%	>99%
Mean birth weight (grams)	3083	400	2.0%	>99%	>99%	>99%	>99%

Trial Registry

- <https://clinicaltrials.gov/ct2/show/NCT01119612>
- Public record, even if study not completed or not published
- Accountability for primary outcomes / prevent fishing for statistical significance

IRBs and Informed Consent

- Institutional Review Board
- Protect the rights, welfare, and privacy of human research subjects
- Multiple IRBs may be involved in a trial
- Before any participants are enrolled or data are collected, the IRB reviews the study protocol, tools (including questionnaires), and informed consent forms
- An IRB can approve, disapprove, monitor, and require modifications in all research activities that fall within its jurisdiction
- Review informed consent form

Data and Safety Monitoring Board (DSMB)

◦ groups of experts not directly involved in the study, but are experts in their respective field in addition to the IRB members.

◦ Additional layer of protection if study has extra potential of risk/harm.

- Responsible for oversight of the **safety and efficacy** of the study
- Address issues of research participant protection
- Meets periodically before and during a trial
- Develop recommendations regarding **protocol modifications** and **continuation** of the clinical trial
 - Based upon observed **beneficial or adverse effects** of the interventions
 - Examine **endpoint and safety data** from the study
 - Review **overall** study procedures and progress, and identify and resolve any issues as needed
 - Can recommend that a trial is **suspended or terminated**

DSMB Sessions

Feature	IRB	DSMB
Primary Focus	Ethics, participant rights, and regulatory compliance	Safety and scientific integrity of clinical trials
Data Review	Generally does not review unblinded data	Independently reviews interim, unblinded data
Action	Approves, requires modifications, or disapproves research	Recommends continuation, modification, or termination of a study
Requirement	Required for all human research	Required for high-risk studies, especially multi-site Phase III trials

- Level 1: Open Sessions – The DSMB will review issues relating to the overall operations of the trial:
 - Accrual and follow-up rates
 - Baseline characteristics
 - Quality control and quality assurance
 - Compliance with the study protocol
 - Follow-up procedures
- Level 2: Closed Safety and Efficacy Sessions – The DSMB will review safety and efficacy data according to coded treatment status

Unblinding and Stopping Rules

- Why would we unblind?
 - Assess harms / benefits attribution
 - If unexpected / severe events occur.
 - Protect participant health / safety.
- Why would we terminate the trial?
 - Safety concerns: harmful effects ↑
 - Efficacy signal: benefits are so strong that it's unethical to withhold Trt from control
- Attribution to study interventions?
 - Futility: trial is unlikely to show difference even if continued.
 - Can be good or bad
 - Must attribute to study before deciding to unblind or stop.
- Safety outcomes
 - Placental malaria
 - Severe maternal anemia (Hb <7 g/dL)
 - Side effects of iron supplementation, particularly severe nausea or vomiting
 - Any maternal hospitalizations
- Efficacy outcomes
 - Fetal loss
 - Prematurity
 - Low birth weight
- Serious Adverse Events (SAEs)
 - Child death
 - Participant (maternal) death
 - Stillbirth
 - (Spontaneous) abortion
 - Birth defect

Unblinding and Stopping Rules

- Safety endpoints
 - If the difference between the two treatment groups is statistically significant at $p < 0.05$, the findings will be flagged for discussion by the DSMB *unethical that Trt. is not given to the Control.*
 - The study may be unblinded, stopped, or terminated if recommended by the board
- Efficacy endpoints
 - Various stopping rules exist *interim*
 - For all interim analyses, the study may be stopped at the discretion of the DSMB if the difference between the treatment groups is statistically significant at $P < 0.001$ *DSMB may stop the trial early to offer the better Trt. for everyone.*
- Final analyses used a significance level of $\alpha = 0.05$
- Final unblinding and stopping rules are at the discretion of the board
- What is the (statistical) risk of doing several interim analyses?

Multiple Testing

- Suppose you are comparing treatment vs. control, and the treatment has zero effect (H_0 is true)

- ^{Type I Error} $P(\text{false positive}) = P(\text{reject } H_0 \mid H_0 \text{ true}) = \alpha =$
significance level

- Correct decision: $P(\text{do not reject } H_0 \mid H_0 \text{ true})$
 $= 1 - \alpha$ ^{true negative}

- If you do one test at $\alpha=0.05$, there is a 5% chance of falsely rejecting H_0 (false positive)

Multiple Testing

- If you do two independent statistical tests:

- Correct decision:

$$\begin{aligned} & P(\text{do not reject either } H_0 \mid \text{both } H_0 \text{ true}) \\ &= P(\text{do not reject } H_{0_1} \mid H_{0_1} \text{ true}) * P(\text{do not reject } H_{0_2} \mid H_{0_2} \text{ true}) \\ &= (1-\alpha) * (1-\alpha) \\ &= (0.95)^2 = 0.9025 \end{aligned}$$

- If $\alpha=0.05$ and you did only 1 test, $P(\text{incorrect decision}) = \alpha = 0.05$
- If you did 2 tests, $P(\text{both correct decisions}) = (1-\alpha)^2 = (0.95)^2 = 0.9025$
 - $P(\text{at least 1 incorrect decision}) = 1 - 0.9025 = 0.0975 \uparrow > 0.05$

Even though each test is only 5% likely to make an error, doing 2 tests gives us almost 10% chance of getting a false positive!

Multiple Testing

- As you do more statistical tests, the chance of having at least one false positive increases

* Interim analyzes + multiple endpoints $\uparrow$ false positive.

($P < 0.001$)

- Remedies
 - Adjust α (Ex: Bonferroni...)
 - Use group sequential designs
 - Apply alpha-spending functions

> why DSMB set stricter early threshold to preserve overall error control.

1. Serious Adverse Events

	Treatment 1			Treatment 2			P-value ^a
	Events	No. Participants	%	Events	No. Participants	%	
Child death	8	207	3.9%	0	208	0.0%	0.004 ^b
Stillbirth	6	213	2.8%	7	215	3.3%	0.79
Abortion	14	430	3.3%	7	437	1.6%	0.11
Birth defect	0	207	0.0%	1	208	0.5%	1.00 ^b

^a Based on a χ^2 test of independence unless otherwise indicated.

^b Based on Fisher's exact test.

Child Deaths in Malaria1

	Treatment 1			Treatment 2			P-value ^a
	Events	No. Participants	%	Events	No. Participants	%	
As of 31 Nov 2011, including questioned case	8	207	3.9%	0	208	0.0%	0.004 ^b
As of 31 Nov 2011, excluding questioned case	7	206	3.4%	0	208	0.0%	0.007 ^b
As of present day, including questioned case	11	329	3.3%	5	325	1.5%	0.14
As of present day, excluding questioned case	10	328	3.0%	5	325	1.5%	0.20

^a Based on Chi-square test of independence unless otherwise indicated.

^b Based on Fisher's exact test.